

Case Study Report

Design and Creative Technologies

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Review Pulse v2: Aspect-Based Sentiment Analysis of Customer Reviews with Attention-Based Deep Learning

1. Abstract

Customer reviews carry mixed opinions - praise for one aspect, criticism of another - yet most sentiment systems, including our Assessment 1 N-gram classifier, reduce a review to a single label. This proposal designs Review Pulse v2, an aspect-based sentiment analysis (ABSA) system that predicts sentiment per aspect of a review. We compare an attention-based LSTM (ATAE-LSTM) with a fine-tuned transformer (DistilBERT) on the SemEval-2014 benchmark and add an attention-visualization layer for interpretability. The proposal states the problem, synthesizes the literature, sets out the method, and presents a feasible plan toward the Assessment 3 build.

2. Problem Statement, Aim and Research Questions

Problem. Most sentiment systems assign a single polarity to a whole text. Our Assessment 1 N-gram classifier and the deep CNN of Zhao, Gui and Zhang (2018) both do exactly this. Real customer reviews are mixed: "*the food was great but the service was slow*" carries two opposite opinions, one per aspect. A single label collapses that detail and hides precisely what product, hospitality, and customer-experience teams need to act on.

Aim. Design and evaluate an aspect-based sentiment analysis system that predicts sentiment per aspect of a review, compares an attention-based LSTM (ATAE-LSTM) with a fine-tuned

transformer (DistilBERT), and provides human-readable attention or attribution visualizations for each prediction.

Research questions.

- RQ1 - How much does explicit aspect conditioning improve sentiment classification on multi-aspect sentences compared with a target-agnostic baseline?
- RQ2 - How does an attention-LSTM compare with a fine-tuned transformer on the SemEval-2014 aspect sentiment task (accuracy, macro-F1)?
- RQ3 - What human-readable evidence do attention or attribution visualizations provide for aspect-level predictions?

3. Literature Review

Deep learning reset the baseline for sentiment analysis: Zhao, Gui and Zhang (2018) show that a convolutional network over word vectors outperforms hand-engineered features for Twitter sentiment. Yet their model, like our Assessment 1 N-gram classifier, resolves a whole text to a single polarity - the very limitation that aspect-based sentiment analysis (ABSA) exists to remove. Pontiki et al. (2014) reframed the problem in SemEval-2014 Task 4, defining the aspect-level benchmark and annotation scheme we adopt and shifting the question from "is this review positive?" to "which aspect is positive?".

The sequence-modelling line answers that question by conditioning on the aspect. Tang et al. (2016) show that target-dependent LSTMs (TD-LSTM) outperform target-agnostic ones by encoding the aspect's position, but they still weight all context words equally. Wang et al. (2016)

close that gap with ATAE-LSTM, adding an aspect embedding and an attention mechanism so the model learns which words matter for which aspect - the design we adopt as our first model and the source of our interpretability layer.

The transformer era diverges from recurrence. Devlin et al. (2019) replace sequential encoding with pretrained bidirectional attention, and Sun, Huang and Qiu (2019) adapt BERT to ABSA by constructing an auxiliary sentence per aspect, recasting classification as sentence-pair inference - the basis of our second (DistilBERT) model. Recent work pushes further: comparative benchmarks report strong transformer results on the same SemEval-2014 sets (Jayakody et al., 2024), and large language models now perform ABSA in zero- and few-shot settings (Simmering & Huoviala, 2023), a shift mapped by a recent systematic review of the field (Hua et al., 2024).

We build on, rather than chase, this frontier. Heavy LLM pipelines rarely pair a light, reproducible model with faithful explanations; and where reviews lack gold aspects, neural aspect discovery (He et al., 2017) can surface them unsupervised - our optional Topic Modelling stage. Our niche is explainable, low-compute aspect sentiment, evaluated honestly against this body of work.

4. Proposed Approach and Methods

Dataset. We use SemEval-2014 Task 4 - Restaurants (~3k sentences) and Laptops (~3k), each annotated with aspect terms and polarity. The data is already annotated, which removes manual labelling and is a key feasibility factor; the restaurant domain maps cleanly to customer-experience framing.

Scope. We focus on aspect sentiment classification given gold aspect terms (the well-defined ATSC setting) rather than full end-to-end aspect extraction, which keeps the Assessment 3 build feasible. Extraction - and an optional neural Topic Modelling stage that discovers aspects when gold labels are absent - are documented stretch goals, not core deliverables.

Model progression. (1) A TF-IDF + logistic-regression sentence-level baseline establishes the aspect gap and ties back to Review Pulse v1. (2) A target-agnostic LSTM tests whether recurrence alone resolves that gap. (3) ATAE-LSTM, an LSTM conditioned on an aspect embedding with aspect-aware attention, tests the value of explicit aspect conditioning. (4) DistilBERT, fine-tuned with the aspect as an auxiliary sentence, is the modern contextual model. (5) An attention or attribution visualisation layer provides indicative, human-readable evidence for each prediction; it is not treated as a causal explanation without additional faithfulness tests.

Evaluation. Accuracy and macro-F1 on aspect sentiment, with a per-class breakdown and a focused analysis of sentences containing conflicting aspect sentiment. Qualitative attention or attribution heatmaps assess interpretability. We use fixed seeds, explicit train/dev/test splits, and guard against data leakage. Cross-domain transfer from Restaurants to Laptops remains an optional extension because their aspect distributions differ.

Deployment. Assessment 3 ships a Streamlit demo where a user types a review and sees per-aspect sentiment plus the attention heatmap, mirroring the live Review Pulse v1 app.

5. Project Plan and Risk Management

The plan runs from this proposal (Module 8) to the Assessment 3 submission (Module 12): data pipeline and classical baseline (Modules 8-9), ATAE-LSTM (Modules 9-10), DistilBERT fine-tuning (Module 10), interpretability and Streamlit demo (Module 11), then evaluation, comparison and report (Modules 11-12), with a submission buffer. Roles are split across implementation, literature, and project management, with equal presentation time. Critical success factors: the data is pre-annotated, compute is light (DistilBERT on free Colab GPU), and a working classical/LSTM path exists even if the transformer underperforms.

Key risks and mitigations: small data risks overfitting (mitigate with dropout, early stopping, and transfer learning); transformer fine-tuning could be heavy (use DistilBERT, contingency ATAE-LSTM only); scope creep (keep extraction and Topic Modelling as gated stretch goals); and attention is not guaranteed to be faithful, so we frame it as indicative, not causal.

6. Conclusion

Sentence-level sentiment loses the aspect-level detail that businesses act on. Review Pulse v2 proposes an explainable, low-compute ABSA system that compares an attention-LSTM with a fine-tuned transformer and visualises its reasoning. The proposal sets a feasible, well-scoped path into the Assessment 3 build.

7. Appendices

7.1. Appendix A – From Assessment 1 to Review Pulse v2

Table A1 – Knowledge transition from Assessment 1 to the proposed Assessment 2 and 3 project.

Assessment 1 - N-gram sentiment	Assessment 2/3 - Review Pulse v2
Count observed word sequences	Learn distributed and contextual representations
Fixed Markov context window	LSTM recurrence, aspect-aware attention and Transformer context
One sentiment label per tweet	One sentiment label per aspect within a review
Add-k smoothing controls sparse counts	Dropout, weight decay, early stopping and transfer learning control overfitting
Hand-defined probability and threshold rule	Learned logits optimized through loss and backpropagation
Inspect bigram probabilities and error examples	Inspect attention or attribution alongside error examples
Accuracy, macro-F1 and confusion matrices	Retain the same metrics, now calculated for aspect-level predictions

End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (Codex-5.5)
- Anthropic Claude (Opus 4.8)

These tools were used to assist with understanding N-Gram probabilistic language models and the Markov assumption, structuring the Python classifier pipeline (preprocessing, bigram counting, add-one smoothing, and the 25% decision rule), debugging code and improving comment quality, interpreting the comparison between the two datasets, improving the clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. *"Explain why add-one (Laplace) smoothing is needed in a bigram language model and how it changes the conditional probability formula."*
2. *"My classifier predicts neutral for very few tweets - how does the 25% threshold rule interact with smoothing to produce that, and how should I explain it critically in the report?"*
3. *"Format this as APA 7th: Go, Bhayani & Huang (2009), Twitter sentiment classification using distant supervision, Stanford CS224N project report."*

I confirm that the use of the AI tool has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

8. References

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